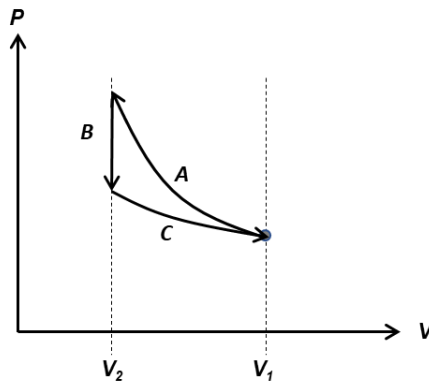


Question	Answer	Marks
2(a)	Arrows drawn Processes are labelled A is a curve (steeper than C at common point) B is a vertical line C is a curve  The diagram is a Pressure-Volume (P-V) graph. The vertical axis is labeled P and the horizontal axis is labeled V. Two vertical dashed lines are drawn at volumes V_2 and V_1, where $V_2 < V_1$. Process A is a curve starting at (V_1, P_1) and ending at (V_2, P_2). Process B is a vertical line segment from (V_2, P_2) to (V_2, P_1). Process C is a curve starting at (V_1, P_1) and ending at (V_2, P_2). Arrows on all three processes indicate a clockwise cycle. At the common point (V_2, P_2), process A is steeper than process C.	A3

2(b)	Isovolumetric/ isochoric	A1
2(c)	Net heat transfer out of cylinder	A1
2(d)	zero	A1
2(e)	Process B and C offers heat transfer out of the cylinder to the bath. This results in the melting of ice in the bath. Net heat transfer out = $(100)(334) = 33400 \text{ J}$ Applying 1st Law of Thermodynamics to process ABC, $\Delta U = Q + W$ For one cycle, there is no internal change in energy. Hence net work done on the gas = $+ 33400 \text{ J}$	M1 A1
	Max Marks	8

Setter: BiaoJin Marker:		
Question	Answer	Marks
2(a)	Since the acceleration of the object can be written in the form $a = \alpha V$ where α is a constant	